Chapter 3

Positive, but not completely

In this chapter we will discuss linear maps which are positive (i.e., they preserve positive semidefiniteness) but not necessarily completely positive. Although such operations cannot directly be implemented in a physical device they play an important role as mathematical tools, in particular in entanglement theory (see Sec.3.2).

Recall that a linear map $T : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_{d'}(\mathbb{C})$ is *positive* if for every positive semidefinite matrix $A \in \mathcal{M}_d(\mathbb{C})$ (we write $A \geq 0$) we have $T(A) \geq 0$. T is positive iff the adjoint map T^* is so, and every positive map is Hermitian in the sense that $T(X^\dagger) = T(X)^\dagger$ for all $X \in \mathcal{M}_d(\mathbb{C})$. By Prop.2.2 every positive map is a difference of two completely positive maps. Since this is, however, true for every Hermitian map it does, unfortunately, not provide a simple parametrization of positive maps.

The paradigm of a map, which is positive but not completely positive, is the usual matrix transposition $A \mapsto \theta(A) := A^T$ which is defined with respect to a given orthogonal basis by $\langle i|A^T|j\rangle = \langle j|A|i\rangle$. The transposition is trace preserving and positive: if $UDU^\dagger$ is the spectral decomposition of any Hermitian matrix (i.e., U is unitary and D diagonal), then $(UDU^\dagger)^T = \bar{U}DU^T$ has the same eigenvalues. The fact that transposition is not completely positive can be easily seen by applying it to half of a maximally entangled state $|\Omega\rangle = \sum_{i=1}^d |ii\rangle/\sqrt{d}$ (see example 1.2):

$$\big(\theta \otimes \mathrm{id}\big)\,\big(|\Omega\rangle\langle\Omega|\big) = \frac{1}{d}\,\mathbb{F}\,, \qquad \mathbb{F} = \sum_{i,j=1}^{d} |ij\rangle\langle ji|\,. \tag{3.1}$$

Since the "flip operator" $\mathbb{F}$ has eigenvalues ± 1 (with multiplicities $d(d \pm 1)/2$ corresponding to symmetric and antisymmetric eigenvectors respectively) θ fails to be completely positive. For the *partial transpose* $(\theta \otimes \mathrm{id})(X)$ we will occasionally write X^{T_A} or X^{T_1} referring to a transposition on Alice's/the first tensor factor.

Transposition plays an important role in the discussion of the structure of positive maps and isometries. We came across one of its appearances already in Wigner's theorem1.1 and we will see similar relations in Prop,3.6 and ... . Among all positive maps transposition gets physically distinguished by a simple interpretation—as time reversal operation (see Sec.3.3).

The simplest class of positive maps are those which are build up of a matrix transposition together with completely positive maps T_1 and T_2. A positive map which has such a decomposition in the form[1]

$$T = T_1 + T_2\theta \tag{3.2}$$

is thus called *decomposable*. Note that the order of maps is irrelevant here, since $\theta T_2 \theta$ is completely positive iff T_2 is. From $\mathcal{M}_d$ to $\mathcal{M}_d$ with $d = 2$ every positive map is decomposable, whereas for larger dimensions this is no longer true. Both facts have far-ranging consequences in entanglement theory (see Sec.3.2). The existence of *indecomposable* positive maps can, with some imagination and the willingness to make a little detour, be traced back to the history of Hilbert's 17'th problem. Since this is a detour of the nicer kind, we will follow it in Sec.3.4. We will start, however, by discussing notions of positivity in between bare positivity and complete positivity:

3.1 n-positivity

A linear map $T : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_{d'}(\mathbb{C})$ is called *n-positive* (with $n \in \mathbb{N}$) if $T \otimes \mathrm{id}_n$ is positive. That is, positivity of T is preserved if we add an 'innocent bystander' of dimension at most n. For given dimensions, the set $\mathcal{T}_n$ of n-positive maps forms a closed convex cone, i.e., if $T_i \in \mathcal{T}_n$ then $\sum_i p_i T_i \in \mathcal{T}_n$ for all positive $p_i \in \mathbb{R}$ and $\mathcal{T}_n$ contains its limit points. Moreover, the dual map T^* is n-positive if and only if T is. Clearly, n-positivity implies m-positivity for all $m \leq n$. Only if $n = d$, then (by Prop.1.2) m-positivity holds for all $m \in \mathbb{N}$, i.e., the map is completely positive. For different n we thus get a chain of inclusions

$$\mathcal{T}_{cp} := \mathcal{T}_d \subseteq \mathcal{T}_{d-1} \subseteq \cdots \subseteq \mathcal{T}_2 \subseteq \mathcal{T}_1, \tag{3.3}$$

from completely positive maps ($n = d$) to positive maps ($n = 1$). As we will see later all these inclusions are strict. To this end the following characterization is useful:

Proposition 3.1 (Choi-Jamiolkowski for n-positive maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_{d'}(\mathbb{C})$ be a linear map and $\tau := (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \in \mathcal{M}_{d'} \otimes \mathcal{M}_d$ its corresponding Choi-Jamiolkowski operator. For $n \leq d$ the following are equivalent:*

1. *T is n-positive.*
2. *$(\mathbb{1} \otimes P)\tau(\mathbb{1} \otimes P) \geq 0$ for all Hermitian projections $P \in \mathcal{M}_d(\mathbb{C})$ with $\mathrm{rank}(P) = n$.*
3. *$\langle\psi|\tau|\psi\rangle \geq 0$ for all $\psi \in \mathbb{C}^{d'} \otimes \mathbb{C}^d$ with Schmidt-rank n.*

[1] Here $T_2\theta$ is a concatenation of maps, i.e., $(T_2\theta)(X) = T_2(X^T)$.

PROOF T is n-positive iff $(T\otimes \mathrm{id}_n)(|\phi\rangle\langle\phi|)\geq 0$ for all $\phi\in\mathbb{C}^d\otimes\mathbb{C}^n$. Embedding $\mathbb{C}^n$ in $\mathbb{C}^d$ this becomes equivalent to $(T\otimes \mathrm{id}_d)(|\phi\rangle\langle\phi|)\geq 0$ for all $\phi\in\mathbb{C}^d\otimes\mathbb{C}^d$ with Schmidt rank n. The latter can be parameterized by $|\phi\rangle = (\mathbb{1}\otimes X)|\Omega\rangle$ where $X\in\mathcal{M}_d$ has rank n. In this way n-positivity becomes equivalent to

$$(\mathbb{1}\otimes X)\tau(\mathbb{1}\otimes X)^\dagger \geq 0, \quad \forall X\in\mathcal{M}_d(\mathbb{C}) : \mathrm{rank}(X)=n. \tag{3.4}$$

From here equivalence with 2. follows by expressing $X=\tilde{X}P$ in terms of the projector P onto its support space and an invertible $\tilde{X}$. Equivalence between 3. and Eq.(3.4) is seen by taking the expectation value of the l.h.s. of (3.4) with an arbitrary vector $|\tilde{\psi}\rangle$ and noting that $(\mathbb{1}\otimes X)^\dagger|\tilde{\psi}\rangle =: |\psi\rangle$ has Schmidt rank n. □

In particular the last item motivates to investigate extremal overlaps with pure states of a given Schmidt rank:

Lemma 3.1 (Overlap with fixed Schmidt rank) *Let $\phi\in\mathbb{C}^{d'}\otimes\mathbb{C}^d$ be a normalized vector with reduced density matrix $\mathrm{tr}_2|\phi\rangle\langle\phi| =: \rho$. Then the maximal overlap with all normalized vectors ψ of Schmidt rank n is given by*

$$\sup_\psi |\langle\phi|\psi\rangle|^2 = ||\rho||_{(n)}, \tag{3.5}$$

where $||\cdot||_{(n)}$ denotes the Ky-Fan norm, i.e., the sum over the n largest singular values (here, eigenvalues).

PROOF Using the Schmidt decomposition of both vectors we can rewrite the optimization in the form

$$\sup_\psi |\langle\phi|\psi\rangle|^2 = \sup_{U,V,\mu}\left|\mathrm{tr}\left[U\sqrt{\mu}V\sqrt{\lambda}\right]\right|^2, \tag{3.6}$$

where U,V are unitaries, λ a diagonal matrix containing the eigenvalues of ρ and $\sqrt{\mu}$ a diagonal matrix containing the Schmidt coefficients of ψ. The maximum in Eq.(3.6) is attained for $U=V=\mathbb{1}$ (which can for instance be seen by exploiting a standard singular value inequality[2]). The remaining supremum is then of the form $\sup_\mu |\sum_{i=1}^n \sqrt{\mu_i\lambda_i}|^2$ with $\{\lambda_1,\ldots,\lambda_n\}$ the n largest eigenvalues of ρ. This can easily be solved using a Lagrange multiplier for the normalization constraint $\sum_{i=1}^n \mu_i = 1$ which gives $\mu_i = c\lambda_i$ with c determined by normalization. Putting things together we arrive at (3.5). □

Utilizing this Lemma together with item 3. in Prop.3.1 we can now provide a simple criterion for n-positivity in terms of the spectral decomposition $\tau = \sum_i \nu_i|\phi_i\rangle\langle\phi_i|$ (with $||\phi_i||=1$). Denoting by ρ_i the reduced density operator of ϕ_i we obtain:

[2]For two matrices A,B the ordered singular values satisfy $\sum_i s_i(AB)\leq\sum_i s_i(A)s_i(B)$; see Thm. IV.2.5 in ... [Bhatia] .

Proposition 3.2 (Criterion for n-positivity) *Consider a Hermitian operator $\tau \in \mathcal{M}_{d'} \otimes \mathcal{M}_d$ (e.g., the Choi-Jamiolkowski operator of a Hermitian map) whose smallest positive non-zero eigenvalue is ν_0. Then with the above notation*

$$\inf_{\psi} \langle\psi|\tau|\psi\rangle \;\geq\; \nu_0 + \sum_{i:\nu_i \leq 0} (\nu_i - \nu_0)||\rho_i||_{(n)}, \tag{3.7}$$

if the infimum is taken over all normalized vectors of Schmidt rank n. Conversely, if all but one eigenvalue ν_- of τ are strictly positive, and ν_∞ is the largest positive eigenvalue, then:

$$\inf_{\psi} \langle\psi|\tau|\psi\rangle \leq \nu_\infty + (\nu_- - \nu_\infty)||\rho_-||_{(n)}. \tag{3.8}$$

Proof By separating positive and negative parts of τ we obtain

$$\tau \;\;\geq\;\; \nu_0 \sum_{i:\nu_i>0} |\phi_i\rangle\langle\phi_i| \;+\; \sum_{j:\nu_j\leq 0} \nu_j |\phi_j\rangle\langle\phi_j|, \tag{3.9}$$

$$\;\;=\;\; \nu_0 \mathbb{1} + \sum_{i:\nu_i\leq 0} (\nu_i - \nu_0)|\phi_i\rangle\langle\phi_i|, \tag{3.10}$$

from which Eq.(3.7) follows by exploiting Lemma 3.1. The inequality (3.9) is reversed if we replace ν_0 by ν_∞, which then leads to Eq.(3.8). □

Now, let us apply this to the map $T_\eta : \mathcal{M}_d \to \mathcal{M}_d$ defined by

$$T_\eta(\rho) = \mathbb{1}_d \mathrm{tr}\,[\rho] - \frac{\rho}{\eta}, \qquad \eta \in \mathbb{R}^+. \tag{3.11}$$

For $\eta \geq d$ this map is completely positive. For $\eta < d$ all positive eigenvalues of the corresponding Choi-Jamiolkowski operator are equal to $1/d$ and there is a single negative eigenvalue $\nu_- = (1/d - 1/\eta)$ with $||\rho_-||_{(n)} = n/d$. Hence, the upper and lower bound in Prop.3.2 actually coincide and we get that T_η is n-positive iff $\eta \geq n$.[3] This proves that all the inclusions in the chain (3.3) are strict. When T_η is positive (i.e., for $\eta \geq 1$) it is decomposable as $T_\eta\theta$ is completely positive.

3.2 Positive maps and entanglement theory

A map, which is positive but not completely positive, like the transposition, does not correspond to a physically implementable operation. Nevertheless, such maps, and in particular the transposition and the map T_1 in Eq.(3.11), have become important mathematical tools in the theory of entanglement. Applied as $(T \otimes \mathrm{id})$ to a bipartite density matrix they are powerful tools for "detecting" entanglement as well as for recognizing "useful" (in the sense of distillable) entanglement.

[3] For the example T_η this can be proven in a simpler way (by averaging a Schmidt rank n vector w.r.t. the group $\{U \otimes \bar{U}\}$), without invoking Prop.3.2. We nevertheless stated the proposition as it can be applied to situations where τ does not have such a symmetry.

Detecting entanglement Suppose that $\rho \in \mathcal{M}_d \otimes \mathcal{M}_{d'}$ is a separable state as defined in Eq.(1.8), then positivity of T implies that $(T \otimes \mathrm{id})(\rho)$ remains positive semi-definite, since tensor products and convex combinations of positive operators are again positive. Hence, for any positive map T a negative eigenvalue of $(T \otimes \mathrm{id})(\rho)$ proves that ρ is entangled:

$$T \text{ positive, and } \langle\Psi|(T \otimes \mathrm{id})(\rho)|\Psi\rangle < 0 \quad \Rightarrow \quad \rho \text{ is entangled.} \tag{3.12}$$

We may go one step further and use an n-positive T. This gives the generalization

$$T\ n\text{-positive, and } \langle\Psi|(T\otimes\mathrm{id})(\rho)|\Psi\rangle < 0 \quad \Rightarrow \quad \begin{array}{l}\text{Every pure state decomposition} \\ \text{of } \rho \text{ contains at least one vector} \\ \text{of Schmidt rank larger than } n.\end{array}$$

The smallest n such that ρ has a convex decomposition into pure states of Schmidt rank at most n is called the *Schmidt number* of ρ. With this terminology separable states are precisely those with Schmidt number one.

In order to prove the converse of the above implications it is useful to introduce the concept of *witnesses* for (Schmidt number $n+1$) entanglement.

Proposition 3.3 (Entanglement witnesses) *A density operator $\rho \in \mathcal{M}_d(\mathbb{C}) \otimes \mathcal{M}_{d'}(C)$ has Schmidt number at least $n+1$ iff there exists a Hermitian operator $W \in \mathcal{M}_d(\mathbb{C}) \otimes \mathcal{M}_{d'}(C)$ such that $\mathrm{tr}\,[W\rho] < 0$ but $\langle\psi|W|\psi\rangle \geq 0$ for all vectors $\psi \in \mathbb{C}^d \otimes \mathbb{C}^{d'}$ with Schmidt rank n.*

Proof Egg–hyperplane–done. In order to be a bit more precise, let us regard the space of Hermitian operators as a real vector space equipped with the Hilbert-Schmidt inner product. Since the set $\mathcal{S}_n$ of density operators with Schmidt number at most n is compact and convex, every $\rho \notin \mathcal{S}_n$ can be separated from it by a hyperplane of Hermitian operators H satisfying $\langle H, \tilde{W}\rangle = c$ for some constant c and some $\tilde{W} = \tilde{W}^\dagger$. That is, $\langle \rho, \tilde{W}\rangle < c$ whereas $\langle \rho', \tilde{W}\rangle \geq c$ for all $\rho' \in \mathcal{S}_n$. Setting $W := \tilde{W} - c\mathbb{1}$ then completes the 'only if' part of the proof. The 'if' part follows from the fact that $\mathcal{S}_n$ is the closure of the convex hull of the set of pure states with Schmidt rank n. □

An operator W with the properties mentioned in Prop.3.3 is called *entanglement witness* (for Schmidt number $n+1$ entanglement). We will follow the literature and use 'entanglement witness' referring to the case $n = 1$ unless otherwise specified. Invoking the Choi-Jamiolkowski correspondence (Prop.2.1) we can assign a positive map to each witness. This leads to the following characterization:

Proposition 3.4 (Positive maps and entanglement) *A density operator $\rho \in \mathcal{M}_d(\mathbb{C}) \otimes \mathcal{M}_{d'}(C)$ has Schmidt number at most n iff for all n-positive maps $T : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_{d'}(\mathbb{C})$ it holds that $(T \otimes \mathrm{id})(\rho) \geq 0$.*

Proof We only have to prove the 'if' part. For that assume that $\rho \notin \mathcal{S}_n$. Then by Prop.3.3 there is an entanglement witness for which $\mathrm{tr}\,[W\rho] < 0$. This defines, by the Choi-Jamiolkowski correspondence, a linear map $T^* : \mathcal{M}_{d'}(\mathbb{C}) \to \mathcal{M}_d(\mathbb{C})$ via

$$W =: (T^* \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|). \tag{3.13}$$

By item 3 in Prop.3.1 T^* and with it its adjoint T is n-positive and by construction

$$\mathrm{tr}\,[W\rho] = \langle\Omega|(T \otimes \mathrm{id})(\rho)|\Omega\rangle < 0. \tag{3.14}$$

Hence, $(T \otimes \mathrm{id})(\rho)$ cannot be positive. □

We see from Eq.(3.14) that the positive map constructed from a witness is actually more powerful, in the sense that it detects more entangled states, than the witness we started with: after all $(T \otimes \mathrm{id})(\rho)$ might have a negative eigenvalue without $\langle\Omega|(T \otimes \mathrm{id})(\rho)|\Omega\rangle$ being negative. A closer look reveals that a positive map T constructed from a witness W 'detects' an entangled state ρ iff it is detected by one of the witnesses from the orbit $\{(\mathbb{1} \otimes X)W(\mathbb{1} \otimes X)^\dagger\}$ with $X \in \mathcal{M}_{d'}(\mathbb{C})$.

Decomposable positive maps correspond to *decomposable witnesses* which are of the form

$$W = P_1 + P_2^{T_1}, \quad \text{with} \quad P_i \geq 0. \tag{3.15}$$

By duality, the existence of indecomposable witnesses for given dimensions is equivalent to the existence of entangled states satisfying $\rho^{T_1} \geq 0$:

Proposition 3.5 (PPT entangled states) [4] *There exists an indecomposable positive map $T : \mathcal{M}_d(\mathbb{C}) \to \mathcal{M}_{d'}(\mathbb{C})$ iff there is an entangled state $\rho \in \mathcal{M}_d \otimes \mathcal{M}_{d'}$ with positive partial transpose $\rho^{T_1} \geq 0$.*

Proof We exploit the equivalence between indecomposable maps and witnesses. That is, we show that the existence of an indecomposable positive map implies that there is an entangled state with $\rho^{T_1} \geq 0$. The converse follows from Prop3.3. Denote by $\mathcal{W} \subseteq \mathcal{M}_d \otimes \mathcal{M}_{d'}$ the set of all entanglement witnesses, i.e., Hermitian operators with $\langle\phi \otimes \psi|W|\phi \otimes \psi\rangle \geq 0$ for all ϕ, ψ, and by $\mathcal{W}_T \subseteq \mathcal{W}$ all decomposable witnesses. Both $\mathcal{W}$ and $\mathcal{W}_T$ are closed convex sets. If there is (by assumption) a $W \in \mathcal{W} \setminus \mathcal{W}_T$ we can separate it by a hyperplane characterized by a Hermitian operator R such that $\mathrm{tr}\,[WR] < 0 \leq \mathrm{tr}\,[W_T R]$ for all $W_T \in \mathcal{W}_T$. Since $\mathcal{W}_T$ contains all positive operators R must be positive so that $\rho := R/\mathrm{tr}\,[R]$ is a density matrix. By construction $\rho^{T_1} \geq 0$ while ρ is entangled due to $\mathrm{tr}\,[W\rho] < 0$. □

We will see shortly that entangled PPT states exists iff the dimension is larger than $\mathbb{C}^2 \otimes \mathbb{C}^3$.

Example 3.1 (Some positive maps and entanglement witnesses)

Transposition *is a positive map. It corresponds to the set of witnesses $W = (A \otimes B)\mathbb{F}(A \otimes B)^\dagger$ for arbitrary A, B. The separability criterion $(\theta \otimes \mathrm{id})(\rho) := \rho^{T_1} \geq 0$ is called* PPT (positive partial transpose) criterion. *Although the partial transpose depends on the choice of the local basis, the PPT criterion does not: if we take the*

[4] Admittedly this proposition has mainly and at most pedagogical value. Once examples of entangled PPT states are constructed it looses a lof of its charm...

partial transposition with respect to a different (local) basis, labeled by the superscript $^{\tilde{T}_1}$*, we get*

$$\rho^{\tilde{T}_1} = (U\otimes\mathbb{1})\big[(U^\dagger\otimes\mathbb{1})\rho(U\otimes\mathbb{1})\big]^{T_1}(U^\dagger\otimes\mathbb{1}) \tag{3.16}$$
$$= \big[(UU^T)\otimes\mathbb{1}\big]\rho^{T_1}\big[(UU^T)^\dagger\otimes\mathbb{1}\big]. \tag{3.17}$$

Eqs.(3.16-3.17) show that the eigenvalues of the partial transpose are basis independent, and so is the positivity of ρ^{T_1}*. Moreover,* $\rho^{T_1}\geq 0$ *is equivalent to* $\rho^{T_2}\geq 0$ *since the two operators only differ by a global transposition. We will see in ... that the PPT criterion is necessary and sufficient for separability in* $\mathbb{C}^2\otimes\mathbb{C}^2$ *and* $\mathbb{C}^2\otimes\mathbb{C}^3$*.*

Reduction criterion. *The set of maps* $T_n(X) := \mathrm{tr}[X]\,\mathbb{1} - X/n$*,* $n\in\mathbb{N}$ *in Eq.(3.11) is n-positive (and self-dual, i.e.,* $T_n^* = T_n$*). As a consequence*

$$n\rho_1\otimes\mathbb{1}\geq\rho \quad \textit{and} \quad n\mathbb{1}\otimes\rho_2\geq\rho, \tag{3.18}$$

(with ρ_k *the k'th reduced density matrix of* ρ*) is necessary for a bipartite state* ρ *to have Schmidt number* n*. For* $n=1$ *this is called* reduction criterion*. As* T_n *is a decomposable positive map, the reduction criterion is generally weaker than the PPT criterion for entanglement detection. For* $\mathbb{C}^2\otimes\mathbb{C}^2$ *the two criteria are equivalent since* $T_1(X) = \big(\sigma_y X\sigma_y\big)^T$ *for any* $X\in\mathcal{M}_2(\mathbb{C})$*. An entanglement witness corresponding to* T_n *is* $W_n = \mathbb{1}/d - |\Omega\rangle\langle\Omega|/n$*. That is, a state* ρ *with Schmidt number* n *has to satisfy* $\langle\Omega|\rho|\Omega\rangle\leq n/d$ *which is a weak version of Lemma 3.1.*

Breuer-Hall map $T_{BH}:\mathcal{M}_d\to\mathcal{M}_d$ *is positive and defined as*

$$T_{BH}(X) = \mathrm{tr}[X]\,\mathbb{1} - X - UX^TU^\dagger, \quad \textit{for any } U \textit{ with } U^T = -U \textit{ and } U^\dagger U\leq\mathbb{1}. \tag{3.19}$$

Positivity is easily seen by applying T_{BH} *to a pure state and noting that due to the anti-symmetry of* U *we have* $\langle\psi|U|\bar\psi\rangle = 0$ *which implies that the subtracted terms are orthogonal. For* d *even there are anti-symmetric unitaries* $U^\dagger = U^{-1} = -\bar U$ *(e.g. embeddings of the Pauli matrix* σ_y*), whereas for* d *odd these do not exist (which can be seen from the fact that* $\det(U)=\det(U^T)=\det(-U)$ *would be equal to* $-\det(U)$ *in this case; see Thms. 3.1,3.2). If* U *is an anti-symmetric unitary* T_{BH} *is not decomposable.*

Choi-type maps. *Let* $D\in\mathcal{B}(\mathcal{M}_d)$ *map a matrix* X *onto a diagonal matrix* $D(X)$ *with the same diagonal entries, and let* U_{k0} *be a cyclic shift as defined in Eq.(2.24). Then for all* $n\in\mathbb{N}$ *with* $1\leq n\leq d-2$ *the map* $T_C\in\mathcal{B}(\mathcal{M}_d)$ *defined by*

$$T_C(X) := (d-n)D(X) - X + \sum_{k=1}^{n} D\big(U_{k0}XU_{k0}^\dagger\big), \tag{3.20}$$

is positive and indecomposable. For $n=0$ *it is completely positive and for* $n=d-1$ *it is proportional to the map* T_n *appearing in the context of the reduction criterion.*

Unextendable product bases *(UPBs) are sets* $S=\{|\alpha_j\rangle\otimes|\beta_j\rangle\}$ *of normalized and mutually orthogonal product vectors in* $C^d\otimes C^{d'}$ *(*$d'\geq d$*) such that there is no product vector, which is orthogonal to every element of* S*, and* $|S| < dd'$*.*[5] *That is, the projection* $P_S := \sum_{j=1}^{|S|}|\alpha_j\rangle\langle\alpha_j|\otimes|\beta_j\rangle\langle\beta_j|$ *has a kernel which contains no product vector. An example of such a UPB in dimension* $3\otimes 3$ *can be constructed from five*

[5] Note that this is only possible if $\{|\alpha_j\rangle\}$ (resp. $\{|\beta_j\rangle\}$) is not a set of mutually orthogonal vectors.

real vectors forming the apex of a regular pentagonal pyramid, where the height h is chosen such that nonadjacent apex vectors are orthogonal. These vectors are

$$v_j = N\left(\cos\frac{2\pi j}{5}, \sin\frac{2\pi j}{5}, h\right), \quad j = 0, \ldots, 4 \, , \tag{3.21}$$

with $N = 2/\sqrt{5+\sqrt{5}}$ and $h = \frac{1}{2}\sqrt{1+\sqrt{5}}$. The UPB is then given by $|\alpha_j\rangle \otimes |\beta_j\rangle = |v_j\rangle \otimes |v_{2j \; mod 5}\rangle$. Since any subset of three vectors on either side spans the full space, there cannot be a product vector orthogonal to all these states. Based on this or any other UPB, one can easily construct an entangled PPT state:

$$\rho_S \propto (\mathbb{1} - P_S). \tag{3.22}$$

Since by construction ρ_S has no product vector in its range it has to be entangled. Moreover, $\rho^{T_1} \geq 0$, since $\{|\bar{\alpha}_j\rangle \otimes |\beta_j\rangle\}$ is again a set of mutually orthogonal and normalized vectors. From here we can construct an entanglement witness

$$W_S = P_S - \epsilon|\psi\rangle\langle\psi|, \tag{3.23}$$

where ψ is any maximally entangled state of dimension d fulfilling $\langle\psi|\rho_S|\psi\rangle > 0$ and $\epsilon := d \inf_{\{\phi_i\}} \langle\phi_1\otimes\phi_2|P|\phi_1\otimes\phi_2\rangle/||\phi_1\otimes\phi_2||^2$. So by construction $\mathrm{tr}\,[W_S\rho_S] < 0$ whereas $\mathrm{tr}\,[W_S\rho] \geq 0$ for all separable states. By the Choi-Jamiolkowski correspondence W_S thus defines a positive map which is not decomposable.

Entanglement distillation ... TO BE WRITTEN ...
LOCC & separable superoperators
PPT implies not distillable
reduction criterion, distillability (maybe majorization)

3.3 Transposition and time reversal

Due to the lack of complete positivity, the transposition (as well as the partial transposition) does not correspond to a physically implementable operation. However, it can physically be interpreted as "time reversal" [**?**] and appears as such quite often as a (global) symmetry, in particular in particle physics [**?**]. Although we will not make explicit use of this interpretation or symmetry in the following, we will briefly describe this relation.

One way to see that the transposition or complex conjugation (which is the same on Hermitian operators) admits an interpretation as time reversal, is looking at the *Wigner function* of a continuous variable system in Schrödinger representation.[6] If we choose $\rho = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i ||\psi_i||^2 = 1$ to be a decomposition of a density operator ρ and define an integral kernel $\mathcal{D}(x;y) = \sum_i \lambda_i \overline{\psi}_i(x)\psi_i(y)$ in the Schrödinger representation, then we can write the Wigner function as

$$\mathcal{W}(x,p) = \pi^{-n} \int \mathcal{D}(q-x; q+x) \, e^{-i2x\cdot p} \, d^n x \, , \tag{3.24}$$

[6] Note that we leave the case $\dim \mathcal{H} < \infty$ within this section. The underlying Hilbert space is now the space $\mathcal{L}^2$ of square integrable functions.

where n is the number of canonical degrees of freedom, i.e., $x, p, q \in \mathbb{R}^n$. From Eq.(3.24) we see, that the complex conjugation or transposition $\mathcal{D}(x;y) \mapsto \mathcal{D}(y;x)$ is equivalent to a reversal of momenta $p \mapsto -p$. Due to the basis dependency of the transposition this is, however, no longer true if we choose another representation. Hence we cannot identify the transposition or complex conjugation with time reversal in general.

Time is a parameter in quantum mechanics. Therefore there cannot be a general transformation in Hilbert space which directly affects this parameter and acts as $t \mapsto -t$. In this sense the term "time reversal" is misleading. It characterizes a reversal of momenta including angular momenta and spins, rather than a reversal of the arrow of time. The defining ansatz for time reversal is thus an operator $\mathcal{T}$ which acts on Hilbert space in a way that position and momentum operators transform as

$$\mathcal{T} Q \mathcal{T}^{-1} = Q \quad \text{and} \quad \mathcal{T} P \mathcal{T}^{-1} = -P, \tag{3.25}$$

and for consistency also $\mathcal{T} L \mathcal{T}^{-1} = -L$ and $\mathcal{T} S \mathcal{T}^{-1} = -S$ for angular momenta and spins. Left and right multiplication of the canonical commutation relations $[Q, P] = i\mathbb{1}$ leads with Eq.(3.25) to $\mathcal{T} i \mathcal{T}^{-1} = -i\,\mathbb{1}$. Thus $\mathcal{T}$ is an "anti-linear" operator, and, as it is supposed to be norm preserving, it is "anti-unitary".[7]

The simplest example of an anti-unitary operator is the complex conjugation Γ (which is on Hermitian matrices equal to the transposition). In fact, in Schrödinger representation we may identify the time reversal operator $\mathcal{T}$ with complex conjugation, since the latter leaves the position operator unchanged and maps $-i\frac{d}{dx} \overset{\Gamma}{\mapsto} i\frac{d}{dx}$ and therefore changes the sign of momentum and angular momentum operators. As the time reversal operator should not depend on the representation (resp. basis), $\mathcal{T}$ cannot be identified with Γ in general. However, every anti-unitary operator can be written as the product of a unitary operator with complex conjugation.

Let us now turn to spin angular momenta. Consider the standard representation, in which S_x and S_z are real and S_y is imaginary. In this basis we have

$$\Gamma S_x \Gamma^{-1} = S_x, \quad \Gamma S_y \Gamma^{-1} = -S_y, \quad \Gamma S_z \Gamma^{-1} = S_z, \tag{3.26}$$

hence we cannot identify $\mathcal{T}$ with complex conjugation. However, we may write $\mathcal{T} = \Gamma V$, where V has to invert the signs of S_x and S_z, while leaving S_y unchanged. Moreover, V is, as a product of two anti-unitaries, a linear unitary operator, and it is supposed to act only on the spin degrees of freedom. These requirements are satisfied by $V = \exp -i\pi S_y$, which rotates the spin through the angle π about the y axis. Hence, we have with respect to the considered representation[8]

$$\mathcal{T} = \Gamma e^{-i\pi S_y}, \tag{3.27}$$

[7] An anti-linear operator A satisfies $Az = \bar{z}A$ for any complex number z. The product of two anti-linear operators is thus again a linear operator. If the inverse A^{-1} exists and $\forall \Psi \in \mathcal{H} : ||A\Psi|| = ||\Psi||$, then we say that A is anti-unitary. We have then $\langle A\Phi | A\Psi \rangle = \langle \Psi | \Phi \rangle$.

[8] Note that time reversal squared is not the identity. The operator $\mathcal{T}^2$ in Eq.(3.27) has eigenvalues -1 for states with total spin $n/2$, with n odd.

and applying the time reversal to a density matrix describing a finite dimensional spin system gives then $\mathcal{T}^{-1}\rho\mathcal{T} = V^\dagger\rho^T V$.

Let us now consider (finite dimensional) composite systems, where $\rho \in \mathcal{B}(\mathcal{H}_1 \otimes \mathcal{H}_2)$. The global time reversal operator is then $\mathcal{T} \otimes \mathcal{T}$, which is a well defined anti-unitary operator again. Unfortunately, there exists no operator on Hilbert space, which corresponds to a "partial time reversal" of only one of the two subsystems. In fact, the action of a tensor product of the form $\mathrm{id} \otimes \mathcal{T}$ (linear$\otimes$anti-linear) on a vector $\Psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$ is not well defined. However, since for product vectors $(\mathrm{id} \otimes \mathcal{T}^{-1})(|\psi\rangle \otimes |\phi\rangle)$ gives only rise to a phase ambiguity, the action on the respective projectors

$$\left(\mathrm{id} \otimes \mathcal{T}^{-1}\right)\left(|\psi\rangle\langle\psi| \otimes |\phi\rangle\langle\phi|\right)\left(\mathrm{id} \otimes \mathcal{T}\right) \tag{3.28}$$

becomes well defined without any ambiguity, and Eq.(3.28) describes a linear operator again. Since the real linear hull of projectors onto product states equals the set of all Hermitian operators, we can in this way define the "partial time reversal" as a map on density matrices. Due to Eq.(3.27) this is equivalent to the partial transposition up to a local unitary transformation. Note that the latter is irrelevant for the detection of entanglement since it preserves the eigenvalues.

The time reversal symmetry of a dynamics, i.e., the invariance of a Hamiltonian under time reversal, does not lead to a conserved quantity (like parity for space inversion). However, it sometimes leads to a degeneracy of the eigenvalues:

Suppose $[H, \mathcal{T}] = 0$ for some anti-unitary $\mathcal{T}$ and Hermitian H. Writing $\mathcal{T} = \Gamma V$ with V unitary this becomes equivalent to $HV^\dagger = V^\dagger H^T$. Moreover, $\mathcal{T}^2 = -\mathbb{1}$ is equivalent to $V^T = -V$ being anti-symmetric.

Theorem 3.1 (Kramer's theorem) *If $[H, \mathcal{T}] = 0$ and $\mathcal{T}^2 = -1$ for a Hermitian H and anti-unitary $\mathcal{T}$, then each eigenvalue of H is at least two-fold degenerate.*

Proof Let $H|\psi\rangle = \lambda|\psi\rangle$. Then $HV^\dagger|\bar\psi\rangle = V^\dagger H^T|\bar\psi\rangle = V^\dagger \bar H|\bar\psi\rangle = \lambda V^\dagger|\bar\psi\rangle$, so $V^\dagger|\bar\psi\rangle$ gives rise to the same eigenvalue. This is degenerate since the two vectors are orthogonal:

$$\langle\psi|V^\dagger|\bar\psi\rangle = -\langle\psi|\bar V|\bar\psi\rangle = -\overline{\langle\bar\psi|V|\psi\rangle} \tag{3.29}$$

$$= -\langle\psi|V^\dagger|\bar\psi\rangle. \tag{3.30}$$

□

Note that anti-symmetric unitaries only exist in even dimensions.[9] In fact, the restriction on the required symmetry in the theorem can be relaxed as the same proof shows:

Theorem 3.2 (Kramer's theorem II) *Let H be Hermitian and $HA = AH^T$ for some anti-symmetric $A \neq 0$. Then each eigenvalue of H is at least two-fold degenerate.*

[9]In odd diemnsions we get that $\det(V) = \det(V^T) = -\det(V)$ which contradicts the fact that the determinant of a unitary has to be a phase.

3.4 From Hilbert's 17th problem to indecomposable maps

The following pages will somehow be a detour following a historical path. Rather than writing down examples of indecomposable positive maps (as we did within example 3.1) we will see how they arose from a problem Hilbert thought about in 1888. For an overview of Hilbert's 17th problem see [?].

Hilbert's problem: A sum of squares (sos) of real polynomials is obviously positive semi-definite. The converse question, whether a real homogeneous psd polynomial of even degree has a sos decomposition, was posed and answered by David Hilbert in 1888. Let $H_h(\mathbb{R}^n)$ be the set of homogeneous polynomials of even degree h in n real variables, and

$$P_h(\mathbb{R}^n) = \{p \in H_h(\mathbb{R}^n) \big| \forall x \in \mathbb{R}^n : p(x) \geq 0\}, \tag{3.31}$$

$$\Sigma_h(\mathbb{R}^n) = \Big\{p \in P_h(\mathbb{R}^n) \big| \exists \{h_k \in H_{h/2}(\mathbb{R}^n)\} : p = \sum_k h_k^2\Big\}, \tag{3.32}$$

the set of psd polynomials in H and the subset of polynomials having an sos decomposition, respectively.[10] It is not difficult to see that $P = \Sigma$ if $n = 2$ or $h = 2$. In fact, the latter is nothing but the spectral decomposition of psd matrices. In 1888 Hilbert showed at first, that every $p \in P_4(\mathbb{R}^3)$ can be written as the sum of squares of three quadratic forms, i.e., $P_4(\mathbb{R}^3) = \Sigma_4(\mathbb{R}^3)$. Moreover, he proved, albeit in a non-constructive way, that $(h, n) = (2, n), (h, 2), (4, 3)$ are in fact the only cases where $P = \Sigma$. In 1900 Hilbert posed a generalization of these questions as his "17th problem" at the International Congress of Mathematics in Paris[11]: Does every homogeneous psd form admit a decomposition into a sum of squares of rational functions? This was answered in the affirmative by Artin in 1927 using the Artin-Schreier theory of real closed fields [?] .[12]

Choi's example: Concerning Hilbert's original 1888 work it took nearly 80 years for explicit polynomials $p \in P \backslash \Sigma$ to appear in the literature. One of these examples was provided by Choi, while investigating one of the numerous "sum of squares" variants of Hilbert's problem. Choi showed in 1975 that there are psd biquadratic forms that cannot be expressed as the sum of squares of bilinear forms. More precisely, let

$$F(x, y) = \sum_{i,j,k,l=1}^{n} F_{ijkl} x_i x_j y_k y_l \ , \quad \forall x, y : F(x, y) \geq 0 \tag{3.33}$$

[10] Note that P and Σ are both convex cones. Moreover, they are closed, i.e., if $p_n \to p$ coefficientwise, and each p_n is in P resp. Σ, then so is p.

[11] At this congress Hilbert outlined 23 major mathematical problems. Whereas some are broad, such as the axiomatization of physics (6th problem), others were very specific and could be solved quickly afterwards.

[12] Artin's proof holds for forms with coefficients from a field with unique order, i.e., in particular for $\mathbb{R}$.

be a real psd biquadratic form in $x, y \in \mathbb{R}^n$. For $n = 3$ Choi found a counterexample to the (wrongly proven[13]) conjecture that there is always a decomposition into a sum of squares of bilinear forms, i.e.

$$F(x,y) \stackrel{?}{=} \sum_\alpha \big(f^{(\alpha)}(x,y)\big)^2 = \sum_\alpha \Big(\sum_{ij=1}^n f_{ij}^{(\alpha)} x_i y_j\Big)^2. \tag{3.34}$$

Choi's counterexample, for which he gave very elementary proofs, is

$$F(x,y) = 2\sum_{i=1}^3 x_i^2(y_i^2 + y_{i+1}^2) - \Big(\sum_{j=1}^3 x_j y_j\Big)^2, \tag{3.35}$$

with $y_{3+1} \equiv y_1$. Note that $F \in P_4(\mathbb{R}^6)$ and that every quadratic form appearing in an sos decomposition of F has to be bilinear. Hence, Eq.(3.35) is an element of $P\backslash\Sigma$ for $(h,n) = (4,6)$.[14]

For the mere beauty of the example let us mention an element of $P\backslash\Sigma$ for $(h,n) = (4,5)$:

$$\sum_{i=1}^5 \prod_{j\neq i}(x_i - x_j). \tag{3.36}$$

Relation to positive maps: In our context the main significance of biquadratic forms lies in their relation to linear maps on symmetric matrices. This relation is in fact the real analogue of the correspondence between entanglement witnesses and positive maps expressed in Eq.(3.13). Let $\mathcal{S}_n$ be the set of all real symmetric $n \times n$ matrices, and let $\Phi : \mathcal{S}_n \to \mathcal{S}_n$ be a positive linear map on $\mathcal{S}_n$, i.e. $\forall x \in \mathbb{R}^n : \Phi\big(|x\rangle\langle x|\big) \geq 0$. Then Φ corresponds to a psd biquadratic form F and vice versa via[15]

$$F(x,y) = \langle y|\Phi\big(|x\rangle\langle x|\big)|y\rangle. \tag{3.37}$$

Particular instances of such maps are *congruence maps* of the form $\Phi(A) = K^T A K$, which in turn correspond to psd biquadratic forms $F(x,y) = f(x,y)^2$, where $f(x,y) = \sum_{ij} K_{ij} x_i y_j$ is a bilinear form. The existence of a psd biquadratic form, which does not admit an sos decomposition, thus implies that the set of positive maps $\Phi : \mathcal{S}_n \to \mathcal{S}_n$ properly contains the convex hull of all congruence maps (for $n \geq 3$). The latter is, however, the real analogue of the set of (complex) decomposable positive maps. Let us now see, whether the map

$$\Phi(S) = 2\big[\mathrm{tr}\,[S]\,\mathbb{1} - \mathrm{diag}(S_{33}, S_{11}, S_{22})\big] - S, \tag{3.38}$$

[13]For the case $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$ with $n = 2$ or $m = 2$ Calderon [?] proved that there is always a sos decomposition of the form in Eq.(3.34). In [?] Koga claimed that this is also true for arbitrary n, m. Finding the flaw in Koga's proof, was apparently Choi's motivation for [?].

[14]By choosing dependent x and y, Choi also specified $F(x,y)$, such that it yielded elements of $P\backslash\Sigma$ for $(h,n) = (4,4)$ and $(6,3)$ respectively.

[15]$F \geq 0$ implies $\Phi \geq 0$ since any symmetric matrix $S \in \mathcal{S}_n$ is psd if $\langle y|S|y\rangle \geq 0$ for every real vector $y \in \mathbb{R}^n$.

which corresponds to the psd biquadratic form in Eq.(3.35) (see also example 3.1), represents an admissible complex non-decomposable positive map as well. Let Λ be a complex extension of Φ defined as in Eq.(3.38) but on arbitrary complex 3×3 matrices. Obviously, Λ maps Hermitian matrices onto Hermitian matrices. Moreover, if Λ is a decomposable positive map, it acts on symmetric matrices as $\Lambda(S)=\sum_j K_j^* S K_j$ for some complex matrices $\{K_j\}$. If we decompose the latter with respect to their real and imaginary parts $K_j = R_j + iI_j$ we get

$$\Lambda(S)=\sum_j R_j^T S R_j + I_j^T S I_j \overset{!}{=} \Phi(S), \tag{3.39}$$

contradicting the fact that Φ in Eq.(3.38) has no such decomposition into congruence maps. Hence, Λ is either not decomposable or not a positive map at all. The latter can, however, be excluded, since we have for every vector $z\in\mathbb{C}^3$ with $z_j = x_j\varphi_j$, $x_j = |z_j|$:

$$\Lambda\big(|z\rangle\langle z|\big) = D^*\Phi\big(|x\rangle\langle x|\big)D \;\geq\; 0\,,\quad D=\mathrm{diag}(\varphi_1,\varphi_2,\varphi_3). \tag{3.40}$$

By Prop.(3.5) Choi's map thus proves, albeit in a non-constructive way, the existence of PPT entangled states in $3\otimes 3$.

The question whether an entanglement witness W is decomposable or not, is equivalent to asking whether the biquadratic Hermitian psd form

$$W(\Phi,\Psi)=\langle\Phi\otimes\Psi|W|\Phi\otimes\Psi\rangle=\sum_{ijkl} W_{ijkl}\bar\Phi_i\bar\Psi_j\Phi_k\Psi_l \tag{3.41}$$

has an sos decomposition. It is therefore the complex analog of Eq.(3.34). Apart from examples there is, however, not much known about forms which do not admit such a decomposition.

Every real unextendable product basis (as the one in Eq.(3.21)) can be used to construct a real-valued non-decomposable entanglement witness. These witnesses thus correspond to biquadratic psd forms, which do not admit an sos decomposition and are therefore elements of $P\backslash\Sigma$.

3.5 Preservation of cones and balls

Proposition 3.6 (Automorphisms and rank preserving maps) *Consider a linear map $T:\mathcal{M}_d(\mathbb{C})\rightarrow\mathcal{M}_d(\mathbb{C})$. The following are equivalent:*

1. *T maps the cone of positive semidefinite matrices onto itself.*
2. *T is positive and preserves the rank of Hermitian matrices.*
3. *There is an invertible $Y\in\mathcal{M}_d(\mathbb{C})$ such that T is either of the form $T(X)=YXY^\dagger$ or $T(X)=YX^TY^\dagger$.*

Proof $1 \to 2$: Note first that T is bijective and T^{-1} is again positive, linear and maps the positive semidefinite cone onto itself. Consider an arbitrary positive semidefinite $P \in \mathcal{M}_d(\mathbb{C})$ and the cone defined by

$$\mathcal{C}(P) := \{A \in \mathcal{M}_d(\mathbb{C}) \mid \exists c > 0 : P \geq cA \geq 0\}. \tag{3.42}$$

Note that the subspace spanned by $\mathcal{C}(P)$ has dimension $\text{rank}(P)^2$ and that, by the properties of T, we have $T(\mathcal{C}(P)) = \mathcal{C}(T(P))$. Since T (as a bijection) preserves the dimension of subspaces, this implies that $\text{rank}(P) = \text{rank}(T(P))$. In order to extend this to Hermitian matrices let us decompose an arbitrary $H = H^\dagger$ into $H = P_+ - P_-$ where $P_\pm$ are positive semidefinite and orthogonal. Then $\text{rank}(T(H)) \leq \text{rank}(T(P_+)) + \text{rank}(T(P_-)) = \text{rank}(H)$. Applying the same to $T^{-1}(H)$ instead of H gives the converse inequality.

$2 \to 3$: Define $\tilde{T}(X) := T(\mathbb{1})^{-1/2} T(X) T(\mathbb{1})^{-1/2}$. Since $T(\mathbb{1})$ has full rank, $\tilde{T}$ inherits the properties of T, i.e., it is positive and rank preserving and in addition unital. Now recall that $\lambda \in \text{spec}(H)$ iff $H - \lambda\mathbb{1}$ has reduced rank. Using that $\tilde{T}(H - \lambda\mathbb{1}) = \tilde{T}(H) - \lambda\mathbb{1}$ together with the fact that $\tilde{T}$ preserves the rank for Hermitian H we get that it actually preserves the spectrum. Then Corr.1.1 completes this part of the proof by setting $Y = U(T(\mathbb{1}))^{-1/2}$. $3 \to 1$ should be obvious. □

From the definition of decomposable positive maps in Eq.(3.2) together with the Kraus decomposition in Thm.2.1 it is clear that every extreme point within the convex set of decomposable positive maps is of the form $X \mapsto YXY^\dagger$ or $X \mapsto YX^TY^\dagger$ (with Y not necessarily invertible). That the converse is true as well, i.e., that all these maps are extreme point has been proven in ... .

Proposition 3.7 (Confining Lorentz cones) *Consider a Hermitian matrix $A \in \mathcal{M}_d(\mathbb{C})$. Then*

1. $A \geq 0 \Rightarrow \text{tr}\,[A]^2 \geq \text{tr}\left[A^2\right]$. *Conversely,*
2. $A \geq 0 \Leftarrow \text{tr}\,[A]^2 \geq (d-1)\text{tr}\left[A^2\right]$.

Proof 1. becomes obvious when writing it out in terms of the eigenvalues $\{a_i \in \mathbb{R}\}$: $(\sum_i a_i)^2 \geq \sum_i a_i^2$ is certainly true for $a_i \geq 0$ as the l.h.s. contains all terms of the r.h.s. plus some extra positive terms.

In order to prove the second implication (by contradiction) assume that A has a negative part such that it admits a non-trivial decomposition of the form $A = A_+ - A_-$ with $A_\pm \geq 0$ and $\text{tr}\,[A_+A_-] = 0$. Denoting by $d_+ < d$ the rank of A_+ we can bound

$$\begin{aligned} \text{tr}\,[A] &\leq \text{tr}\,[A_+] = ||A_+||_1 \leq \sqrt{d_+}||A_+||_2 && (3.43)\\ &\leq \sqrt{d-1}||A_+||_2 < \sqrt{d-1}||A||_2, && (3.44)\end{aligned}$$

where the first inequality is a simple consequence of Hölder's inequality[16] and the last one uses $||A||_2^2 = ||A_+||_2^2 + ||A_-||_2^2$ and $A_- \neq 0$. □

[16] For every unitarily invariant norm we have $||CB|| \leq ||\ |C|^p||^{1/p}||\ |B|^q||^{1/q}$ for all $p > 1$ and $1/p + 1/q = 1$. Eq.(3.43) follows from applying this to the trace norm with $p = 2$, $C = A_+$ and B the projection onto the range of A_+.

Lemma 3.2 (Making positive maps trace preserving) *Let* $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ *be a positive map for which* $T^*(\mathbb{1}) > 0$. *There exists an invertible* $X \in \mathcal{M}_d(\mathbb{C})$ *such that* $T'(\rho) := T(X\rho X^\dagger)$ *is a trace-preserving positive map.*

Proof Let $\tau := (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)$ be the Choi-Jamiolkowski operator corresponding to T and τ' the one for T'. By Prop.2.1 the map T' is trace preserving if the partial trace τ'_B equals $\mathbb{1}/d$. Since (again by Prop.2.1) $\tau'_B = X^T \tau_B \bar{X} = X^T T^*(\mathbb{1})^T \bar{X}/d$ this can be achieved by $X := \sqrt{T^*(\mathbb{1})/d}$. □

Hildebrand: ... maps preserving the Lorentz cone ...

Gorini, Sudarshan: ... maps preserving the sphere + Kossakowski and Chruscinski ...

3.6 Complexity issues

3.7 Literature